The Role of Barotropic versus Baroclinic Feedbacks on the Eddy Response to Annular Mode Zonal Wind Anomalies

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ABSTRACT: Multiple independent estimates of the relative role baroclinic versus barotropic processes on the eddy response to EOF1 zonal wind anomalies are developed. All methods conclude that barotropic mechanisms are most important, however, baroclinic mechanisms are not negligible. Additional experiments with the baroclinic feedback turned off (via manipulations to the vertical friction profile) also suggest that barotropic feedbacks are dominant. The methods for estimating the feedbacks are: 1) Rossby Wave Chromatography, 2) forced manipulations of the vertical structure of EOF1 using Linear Response Functions and 3) quantitatively inferring the meridional wave propagation from the mean wave activity budget and then using this to analyze the wave activity response to anomalies. The last method is also applied to both Northern and Southern Hemisphere observations and similar conclusions regarding the feedbacks are reached.

1. Introduction

Zonal-mean zonal wind anomalies (U) in the extratropics are forced by stochastic eddy momentum flux convergence (EMFC) anomalies in the upper troposphere. In turn, the U anomalies feedback on the EMFC in such a way to reinforce the U anomaly (Robinson (1991); Robinson (1994); Robinson (1996); Lorenz and Hartmann (2001); Lorenz and Hartmann (2003)) and/or cause the U anomaly to propagate poleward/equatorward (James and Dodd (1996); Lee et al. (2007); Sparrow et al. (2009); Chemke and Kaspi (2015); Sheshadri and Plumb (2017)). The positive feedback is typically restricted to EOF1 of U and there is substantial debate whether the positive EMFC/U feedback is due to "baroclinic" or "barotropic" mechanisms. For the baroclinic mechanism, it is hypothesized that U anomalies cause colocated anomalies of baroclinic instability of the same sign. The positive baroclinic instability anomalies are equivalent to increases in the upward flux of wave activity into the upper troposphere (Edmon et al. (1980)). Baroclinic mechanisms argue that increases in the baroclinic wave source leads to increases in the wave activity diverging meridionally at upper levels, which is equivalent to reinforcing EMFC at upper levels (Edmon et al. (1980)). Papers that advocate a baroclinic mechanism include Robinson (1996), Robinson (2000), Lorenz and Hartmann (2001), Chen and Plumb (2009), Rivière (2009), Zhang et al. (2012) and Zurita-Gotor et al. (2014).

For barotropic mechanisms, it is hypothesized that the U anomalies change the meridional wave propagation (i.e. eddy momentum flux) directly, independent of changes in the baroclinic wave source. This could come about through linear or nonlinear mechanisms. Papers that advocate a barotropic mechanism include Hartmann (1995), Hartmann and Zuercher (1998), Jin et al. (2006a), Jin et al. (2006b), Chen and Zurita-Gotor (2008), Barnes et al. (2010), Barnes and Hartmann (2011) and Lorenz (2014b).

In most cases, baroclinic mechanisms have been advocated using the anomalous upper-level wave activity budget. If the convergence of the vertical EP flux in the upper troposphere is greater than or equal to the divergence of the horizontal EP flux, then this is considered sufficient evidence that baroclinic feedbacks are operating. The problem with this approach is that the ratio of the vertical to horizontal EP flux of the *mean* state is not taken into account (Robert et al. (2017)). In the mean state, the vertical EP flux is typically at least several times greater than the horizontal EP flux (e.g. Edmon et al. (1980)), which implies that most baroclinic wave activity does not propagate

meridionally, instead the wave activity dissipates locally. If anomalies behave like the mean state, then the anomalous baroclinic wave source must be significantly larger than the divergence of the horizontal EP flux (i.e. EMFC) for the baroclinic feedback to fully account for the feedback. Robert et al. (2017) first advocated these ideas and coined the term "radiation efficiency", which is the ratio of horizontal EP flux divergence to vertical EP flux convergence in the upper troposphere in the mean state, to better quantify the baroclinic feedback.

Alternatively, Hassanzadeh and Kuang (2019) advocated baroclinic mechanisms by manipulating the vertical structure of EOF1 and observing the eddy response. When EOF1 is made barotropic, the EMFC no longer reinforces the U anomalies, which is consistent with a baroclinic feedback. However, the barotropic U anomalies do not give zero baroclinic source anomalies (see heat flux response in Hassanzadeh and Kuang (2019)), instead the baroclinic source anomalies are strong but in the opposite direction of the standard EOF1 case. In this paper, we take the view that both barotropic and baroclinic mechanisms are operating. With two mechanisms and two experiments (standard EOF and barotropic EOF), one can linearly disentangle the contribution of each mechanism. Repeating the Hassanzadeh and Kuang (2019) under this more general framework, we find that the barotropic mechanism is most important but that the baroclinic feedback is non-negligible.

Unlike all studies advocating baroclinic mechanisms, some studies using barotropic methods go beyond simply diagnosing the source of the feedbacks. They also explicitly predict either the structure of of the leading mode (Jin et al. (2006a); Jin et al. (2006b)) or the latitudinal structure of positive and/or propagating eddy feedbacks (Lorenz (2014a); Lorenz (2014b); Lorenz (2015)). The former studies predict annular mode structure from the mean state and the space-time structure of the transient eddy field. The latter studies predict the eddy momentum fluxes from the mean state and the phase-speed/latitude structure of the baroclinic wave source (convergence of the vertical EP flux in the upper troposphere). Lorenz (2015) called their method Rossby Wave Chromatography (RWC) (Held and Phillips (1987)). All of these studies assume linear dynamics for the eddy field.

The goal of this paper is to develop several independent estimates of the relative contribution of barotropic and baroclinic feedbacks to EOF1 U anomalies. First we provide an experiment which demonstrates that a naive analysis of the upper level wave activity budget is misleading by turning off the baroclinic feedback via modifications to the Rayleigh friction profile. Next, we estimate

the barotropic and baroclinic feedbacks using RWC (Lorenz (2015)) and an updated baroclinic RWC that is presented below. Then we provide alternative estimates of the feedbacks using the Linear Response Function (LRF) method of Hassanzadeh and Kuang (2016) and Hassanzadeh and Kuang (2019). Next we develop a simple model that infers the amount of wave propagation from the time-mean wave activity budget. Applying this model to anomalies provides another estimate of barotropic and baroclinic feedbacks. Finally the inferred propagation simple model is applied to observations to estimate feedbacks. All methods estimate that barotropic feedbacks are more important than baroclinic feedbacks for the GCM (best guess: 70% barotropic and 30% baroclinic). For observations, the estimates are seasonally dependent with barotropic feedbacks ranging from 60% to 100% of the total feedback.

2. Methods

a. GCMs

The multi-level GCM is a standard primitive equation spectral model integrating the vorticity, divergence, temperature and the log surface pressure. The sigma coordinate vertical differencing scheme of Simmons and Burridge (1981) is used. An 8th order hyperdiffusion with a time scale of 0.1 days for the smallest scale waves is applied to the model variables. The only non-typical aspect of the model is the time differencing, which is the AB3-AI2 method of Durran and Blossey (2012). The resolution of all simulations is T85 with 20 equally spaced vertical levels. The control simulation is forced with the diabatic heating and frictional damping of Held and Suarez (1994). The model is run for 6500 days and the first 500 days are discarded to allow for model spin-up.

For efficient implementation of the many long-term-integrations required for Linear Response Function (LRF) analysis (Hassanzadeh and Kuang (2016)) and for the two-level baroclinic Rossby Wave Chromatography (see below), we also use a two-level primitive equation model (GCM) based on Hendon and Hartmann (1985). The prognostic variables of vorticity, divergence and potential temperature are defined on two presssure levels (250 and 750mb). In the equations, the vertically integrated divergence is constrained to be zero for consistency with the assumption of constant surface pressure at the lower boundary. For thermal forcing we use Held and Suarez (1994) and for mechanical damping we use Rayleigh friction at the lower-level with an e-folding time scale of 3 days. In the past, this model has been run at coarse resolution (R15) with semi-implicit time

differencing (Hendon and Hartmann (1985); Robinson (1991)). At the T42 resolution used here, however, the speed of the winds is approximately equal to the phase speed of the fastest gravity mode (which is slower than usual due to the constant surface pressure) and therefore our version is fully explicit with third order Adams-Bashforth time differencing (Durran (1991)). The model is run for 251000 days and the first 1000 days are discarded to allow for model spin-up.

The calculation of EP fluxes (Edmon et al. (1980)) requires more care for a coarse vertical-resolution primitive equation model. To be most consistent with the analogous two-level quasi-geostrophic model, the vertical EP flux convergence at the upper-level (i.e. the baroclinic wave source) of the two-level primitive equation model is:

$$\text{source} = \frac{\cos\phi f \overline{\theta' v'}}{\frac{1}{2}(\overline{\theta}_1 - \overline{\theta}_2)}, \tag{1}$$

where ϕ is the latitude, f is the Coriolis parameter, θ' is the eddy component of $\frac{\theta_1+\theta_2}{2}$, v' is the eddy component of $\frac{v_1+v_2}{2}$, level 1 is the upper level, level 2 is the lower level and the $\frac{1}{2}$ factor in the denominator makes the EP fluxes consistent with the vertical differencing in the two-level primitive equation model.

b. Phase Speed/Latitude Spectra

Space-time cross-spectral analyses of eddy fluxes are calculated using the method of Randel and Held (1991). The temporal spectral analysis is performed over 64 day chunks (= 192 times given the 8 hour sampling time) that overlap by 32 days. The resulting frequency spectrum is further smoothed with a running mean over five adjacent frequency bands. The phase speed spectra are given in terms of angular phase speed multiplied by $a\cos(45°)$, where a is the radius of the earth so that the angular phase speed is approximately equal to the phase speed in the mid-latitude. Typically, the stationary phase speed/frequency is not considered because the effects of strong stationary wave sources dominate the observed phase speed spectra. In our models with axisymmetric boundary conditions, however, the stationary frequency (phase speed) is not enhanced relative to adjacent frequency bands and we therefore consider the full phase speed spectrum.

c. *Rossby Wave Chromatography (RWC)*

1) Barotropic RWC

Barotropic RWC (Lorenz (2015)) is based on the barotropic vorticity equation on a sphere linearized about a background zonal-mean zonal wind, $\bar{u}$, and absolute vorticity gradient, β^* $(= a^{-1}\partial_\phi(f+\bar{\zeta}))$,

$$\frac{\partial \zeta'}{\partial t} + \frac{\bar{u}}{a\cos\phi}\frac{\partial \zeta'}{\partial \lambda} + \beta^* v' - \nu\nabla^2\zeta' = F', \tag{2}$$

where ζ', v' and F' are the eddy relative vorticity, meridional velocity and vorticity forcing, λ is the longitude, a is the radius of the earth, ν is the diffusion coefficient. The forcing, F', represents the input of wave activity via baroclinic instability and the diffusive damping is a parameterization of wave dissipation via nonlinear wave breaking. Forced barotropic and shallow water models have been used before to understand zonal-mean zonal wind variability and change (Vallis et al. (2004); Chen et al. (2007); Barnes et al. (2010); Barnes and Hartmann (2011); Kidston and Vallis (2012)). Unlike these studies, which specify the forcing directly, RWC prescribes the wave activity source and then determine the forcing, vorticity and all other fields from the wave source under the assumption that the forcing and vorticity are related by the linearized barotropic vorticity equation (Lorenz (2015)). The advantages of prescribing the wave source are discussed in Lorenz (2014a); Lorenz (2014b). As in Lorenz (2015), we use the convergence of vertical EP flux (Edmon et al. (1980)) in the upper troposphere for the "baroclinic" wave source and, because the background flow is steady and zonally symmetric, RWC is applied separately to each zonal wavenumber and phase speed (Randel and Held (1991)). The model first predicts the eddy vorticity field, and which then allows the eddy vorticity and momentum fluxes to be calculated.

With RWC, the baroclinic wave activity source and the background flow are decoupled so that each can be manipulated separately to better understand the dynamics. Here, the baroclinic feedback is defined as the response of UV to changes in the amplitude of the baroclinic wave activity source integrated over phase speed. The barotropic feedback is defined as the response to changes in the background flow *and* changes in the phase speed of the baroclinic wave source. The phase speed of the wave source is included with the barotropic mechanism because it is fundamentally coupled to the changes in advection by the background U. Acting in isolation, the phase speed effect

damps U anomalies (Lorenz (2014a); Lorenz (2014b)), so including the phase speed effect with the background flow weakens the barotropic feedback and strengthens the baroclinic feedback.

Let $S_0(c)$ be the distribution of the wave activity source in the control simulation as a function of phase speed, c. The portion of the new source coming from the baroclinic feedback is defined as S_0 scaled by the ratio of the new to control S amplitude:

$$\frac{\int_{-\infty}^{\infty} S_1(c)dc}{\int_{-\infty}^{\infty} S_0(c)dc} S_0(c), \tag{3}$$

where $S_1(c)$ is the wave activity source in the new simulation. The source for the phase speed change is defined as

$$\frac{\int_{-\infty}^{\infty} S_0(c)dc}{\int_{-\infty}^{\infty} S_1(c)dc} S_1(c), \tag{4}$$

which is the new source scaled to have the same integrated amplitude as the original source.

2) Baroclinic RWC

Here the barotropic RWC is extended to a linearized baroclinic model:

$$\begin{aligned} \frac{\partial q_1'}{\partial t} + \frac{\bar{u}_1}{a\cos\phi}\frac{\partial q_1'}{\partial \lambda} + \gamma v_1' - k_T k_D^2 (\psi_1 - \psi_2) &= F_1' + \nu \nabla^2 q_1', \\ \frac{\partial q_2'}{\partial t} + \frac{\bar{u}_2}{a\cos\phi}\frac{\partial q_2'}{\partial \lambda} + \gamma v_2' - k_T k_D^2 (\psi_2 - \psi_1) &= F_2' + \nu \nabla^2 q_2', \end{aligned} \tag{5}$$

where q is the Quasi-Geostrophic (QG) Potential Vorticity (PV) ($q_j = f + \nabla^2\psi_j - k_D^2\left(\psi_j - \psi_{3-j}\right)$ where ψ is the streamfunction), k_D is the inverse deformation radius, γ_j is the zonal-mean meridional PV gradient ($= a^{-1}\partial_\phi q_j$), k_T and k_M are the thermal and mechanical damping rates, level 1 is the upper level and level 2 is the lower level. The inverse deformation radius is given by:

$$k_D^2 = \frac{f^2}{\frac{1}{2}c_p\left(\left(\frac{p_2}{p_0}\right)^\kappa - \left(\frac{p_1}{p_0}\right)^\kappa\right)(\overline{\theta}_1 - \overline{\theta}_2)}, \tag{6}$$

where c_p is the specific heat at constant pressure, p is the pressure, $\kappa = 2/7$, $\overline{\theta}$ is the zonal-mean potential temperature from the GCM, and the $\frac{1}{2}$ factor in the denominator makes the QG model consistent with the vertical differencing in the two-level primitive equation model. The values of

k_T and k_M are taken from the GCM simulation we are attempting to model. In the baroclinic case, $F_j + \nu\nabla^2 q_j$ parameterizes nonlinearities as forcing plus damping, which is like stochastic models of baroclinic turbulence (DelSole (2004)). Unlike most stochastic models, however, F does not have a simple *a priori* distribution (i.e. white noise), instead the forcing is calculated such that it gives a particular vertical EP flux. Like the barotropic RWC, we prescribe the vertical EP flux at each zonal wavenumber and phase speed so that is matches the GCM (for details, see Appendix). The diffusivity, ν, is the one free parameter. We use $\nu = 2.47 \times 10^6$ m^2/s, which minimizes the error in the eddy momentum flux integrated over phase speed and zonal wavenumber. This value of diffusivity is also sufficient to stabilize the system to baroclinic instability for all phase speeds and wavenumbers.

d. Linear Response Function (LRF) Analysis

Hassanzadeh and Kuang (2016) developed a novel method to force U perturbations with arbitrary latitude/pressure structure in a GCM. Using this method, the mean flow can be manipulated in various ways to better understand the dynamics (Hassanzadeh and Kuang (2019)). The methodology first involves running a large number of simulations, each with a prescribed forcing that varies in location (latitude and pressure) across the simulations. This exercise creates a linear operator (perturbations are assumed to be small enough) converting zonal-mean U and temperature forcing to U and temperature. Hassanzadeh and Kuang (2016) then implemented the method in Kuang (2010) to filter this ill-conditioned linear operator so that it can be inverted to find the forcing that creates an arbitrary U and temperature mean state.

We repeat this method in the two-level primitive equation model. All parameters and forcing profiles are the same as Hassanzadeh and Kuang (2016) except for the pressure dimension: instead of applying a Gaussian shaped forcing in pressure, we simply apply the forcing in either the upper or lower level. Our forcing experiments are run for 251000 days at T42 resolution (see section 2a).

3. Results

a. Upper Level Friction

The time mean zonal-mean zonal winds (U) with the control Held and Suarez (1994) parameters has a mid-latitude jet centered at about 42° latitude (Fig. 1a). Next, we calculate EOF1 of the

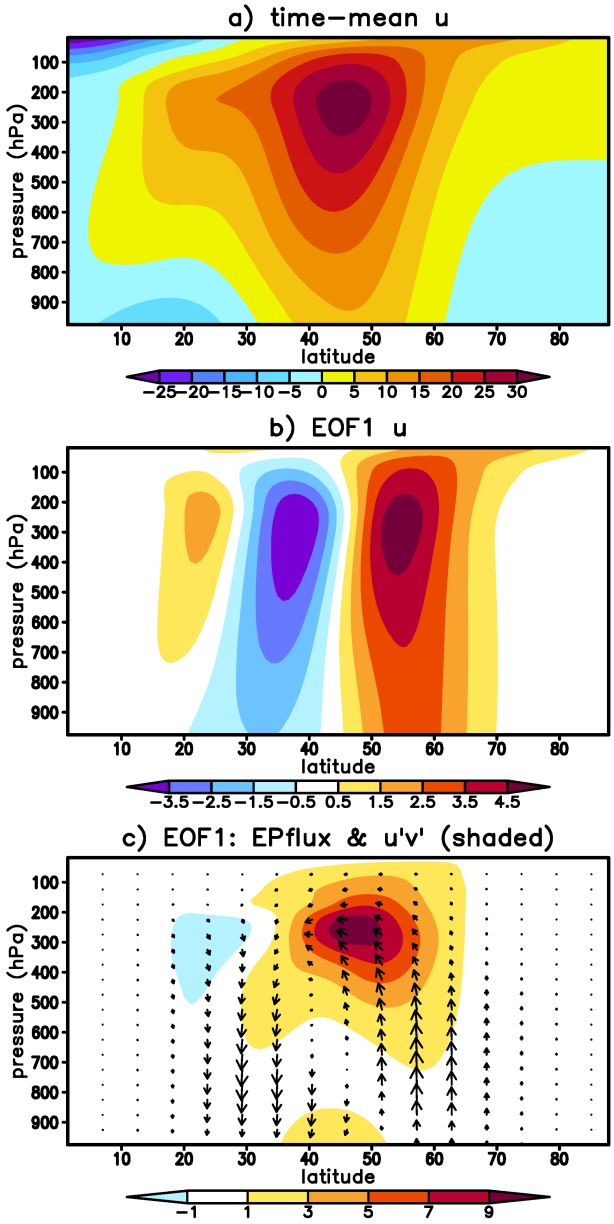

FIG. 1. a) Time- and zonal-mean zonal wind (U) from the control simulation (m/s). b) U anomalies regressed on PC1 of instantaneous U variability (m/s) c) $\overline{u'v'}$ (shaded, m^2/s^2) and EP flux (arrows) anomalies regressed on PC1 of instantaneous U variability.

instantaneous vertical- and zonal-mean zonal wind and then regress the zonal-mean wind on the resulting Principal Component (PC) (Fig. 1b). EOF1 has oppositely signed center of actions on either side of the jet maximum and therefore represents north/south shifts in the mid-latitude jet. In Fig. 1c, the lagged regression between U and UV, and U and the EP fluxes (Edmon et al.

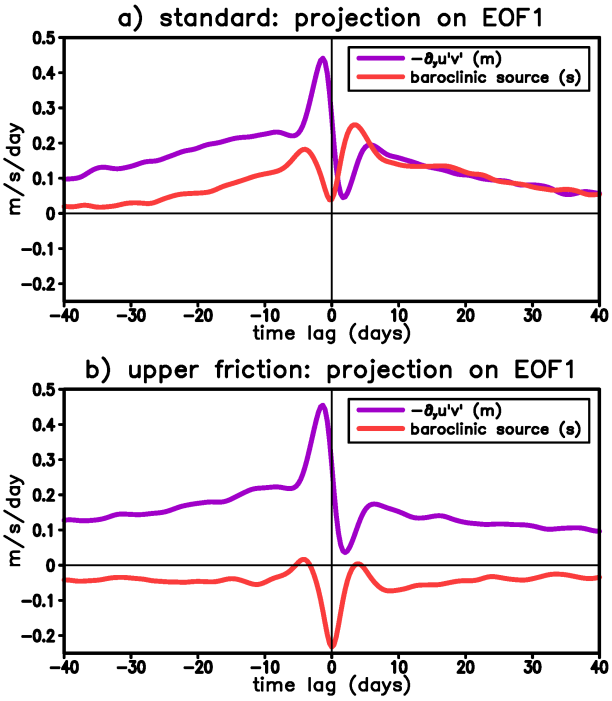

FIG. 2. Lagged regression of upper-troposphere wave activity budget on PC1 of U. Positive lags mean that U leads in time. a) Eddy momentum flux convergence (purple) and baroclinic source (red, convergence of the vertical component of the EP flux in the upper troposphere, see text). b) Same as (a) but for the experiment where friction acts at upper-levels for U anomalies (see text).

(1980)) is averaged from 10 to 20 days *after* the peak in U. These eddy fields are lagged in time to isolate the component of the eddies that is caused by the U anomalies from the stochastic eddy fluxes that generated the U anomalies in the first place (Lorenz and Hartmann (2001)). The UV anomalies "caused" by EOF1 are positive in the upper troposphere at latitudes between the two centers of action of EOF1. This leads to convergence of momentum at the positive center of EOF1 and divergence of momentum at the negative center, which reinforces the EOF1 U anomalies. The horizontal component of the EP fluxes is simply $\cos(\phi)$ times the UV itself; the vertical component of the EP fluxes are proportional to the eddy heat fluxes that generate the waves in the first place. In Fig. 1c, the vertical EP flux anomalies are larger than the horizontal EP fluxes, which suggests that anomalies in baroclinic wave generation are causing the UV anomalies. In the literature, this is called the baroclinic feedback mechanism.

In Fig. 2, the above diagnostics are quantified more precisely by calculating the EP flux budget of the layer from 500 to 100mb. First, define m to be the eddy momentum flux convergence $(-\frac{1}{\cos^2\phi}\frac{\partial\cos^2\phi\overline{u'v'}}{\partial\phi})$ integrated from 500 to 100mb and s to be the convergence of the vertical EP flux integrated from 500 to 100mb. Next, the lagged regression with the U PC is calculated. Finally, the m and s anomalies are projected onto the U EOF1 pattern. The m projection (purple line, Fig. 2a) peaks before the U anomalies (negative lags), which is consistent with the fact that m forces U. The focus here is on the reinforcing m at positive lags beyond 7 days, which is consistent with a positive feedback (Lorenz and Hartmann (2001)). The baroclinic wave activity source, s, anomalies (red) attain the same values as m at positive lags beyond 7 days, which, consistent with the previous figure, suggests that the baroclinic feedback is responsible for the UV anomalies. In the remainder of this paper, however, we show that the baroclinic feedback is not the main driver of UV anomalies. The issue, as described in more detail below, is that reasoning based on the anomalous EP flux budget alone is incorrect. Instead the relative sizes of the vertical and horizontal EP fluxes in the mean state must also be taken into account (see section 3d). First, we describe an experiment which demonstrates that s is not primarily responsible for m.

In addition to UV reinforcing U anomalies, the baroclinic feedback also requires the eddy fluxes to maintain temperature gradient anomalies that are colocated with U anomalies so that baroclinic wave source anomalies are also colocated with U. Robinson (2000) describes in detail how the shape of anomalous upper-level EP flux divergence acts to maintain such temperature gradient anomalies. This theory requires friction at low levels only. Therefore, if the Held and Suarez (1994) friction is moved to upper levels then the U anomalies are no longer be associated with temperature gradient anomalies, which will presumably turn off the baroclinic mechanism. Completely changing the friction from lower to upper levels will dramatically change the mean circulation making the results hard to interpret. Therefore, we only change friction acting on zonal-mean U anomalies:

$$\text{friction} = k_M\left(u' + \overline{\overline{u}}\right) + k_U\left(\overline{u} - \overline{\overline{u}}\right), \tag{7}$$

where the double bar, $\overline{\overline{u}}$, means a time- and zonal-mean, k_M is the standard low-level Held and Suarez (1994) friction profile and k_U is an alternative friction profile acting at upper levels. Here

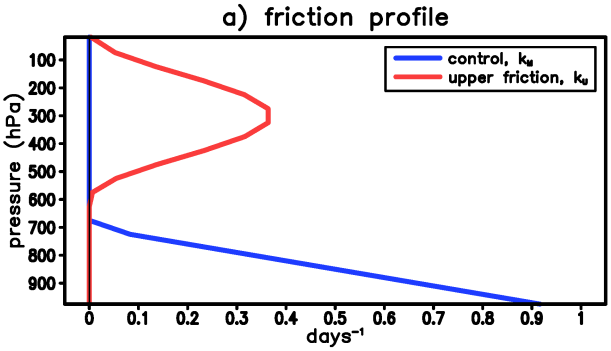

FIG. 3. Friction Rayleigh damping profile for standard Held and Suarez (1994) forcing (blue) and for upper-level friction (red), which acts on U anomalies only.

the k_U profile is given by

$$k_U = \begin{cases} 0.37\text{day}^{-1}\cos^2\left(\pi\frac{p-300mb}{600mb}\right), & \text{if} p < 600mb \\ 0, & \text{otherwise,} \end{cases} \tag{8}$$

where the factor 0.37 gives a vertically-integrated friction time scale (calculated as in Lorenz and Hartmann (2001)) comparable to the control simulation. The friction profiles are plotted in Fig. 3.

In Fig. 4, the results in Fig. 1 are repeated for the upper friction on U anomalies simulation. The time-mean U is practically unchanged from before. The leading EOF has very similar structure in latitude, but the vertical structure is much more barotropic, although not completely so (Fig. 4b). The UV anomalies (Fig. 4c) are the same amplitude as the control simulation, but the vertical EP fluxes in the new simulation have disappeared. Evidently, the UV anomalies do not require baroclinic source anomalies to maintain themselves. In fact, when quantifying the baroclinic source averaged over 500 - 100mb (s, Fig. 2b), the s projection on EOF1 is actually slightly negative. The m projection is essentially unchanged although the persistence of the m anomalies in time is actually enhanced. This experiment shows that m anomalies can be maintained without any s anomalies. Below, we will return to a similar upper friction experiment in the two-level model to make quantitative estimates of the baroclinic versus barotropic feedback.

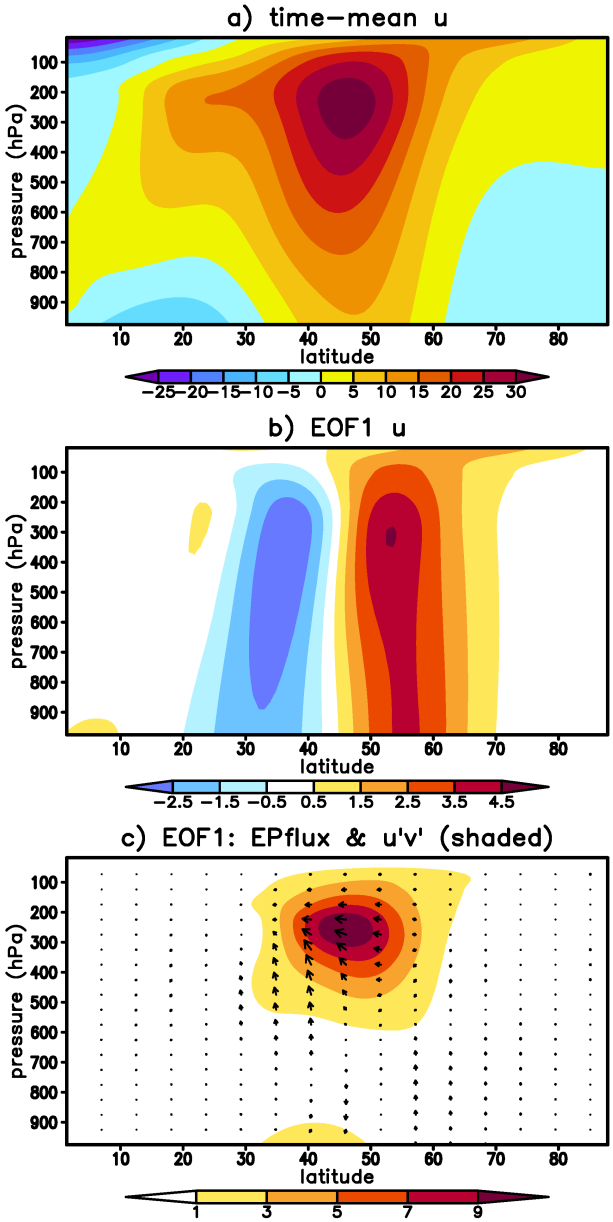

FIG. 4. Same as Fig. 1 but for the simulation with upper-level friction on U anomalies.

b. Rossby Wave Chromatography (RWC)

The second line of evidence suggesting the baroclinic feedback is not dominant comes from RWC experiments. In RWC, the baroclinic wave source and background flow are prescribed independently in a linearized model and therefore the role of each can be separately quantified. As discussed above (Section 2c), the change in phase speed of the wave sources is included in the background-flow/barotropic mechanism. Because the phase speed effect damps most U anomalies

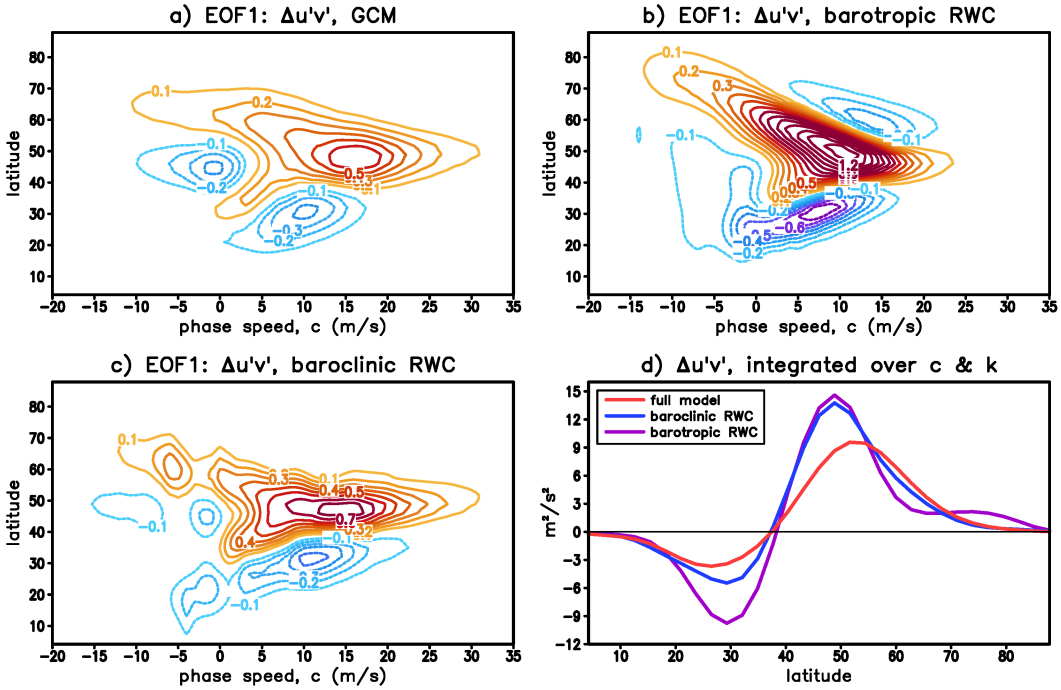

FIG. 5. Comparison of the $\overline{u'v'}$ response to EOF1 in the GCM and two versions of RWC. a) GCM phase-speed/latitude spectrum integrated over zonal wavenumber. b) Barotropic RWC phase-speed/latitude spectrum integrated over zonal wavenumber. c) Baroclinic RWC phase-speed/latitude spectrum integrated over zonal wavenumber. d) $\overline{u'v'}$ response in previous panels integrated over phase speed: GCM (red), barotropic RWC (purple), baroclinic RWC (blue).

(Lorenz (2014a); Lorenz (2014b)), not including this phase speed effect in the baroclinic mechanism actually strengthens the argument that the baroclinic mechanism is not as important.

Because EOF's are by definition transient, we force a permanent EOF1 state in the two-level GCM using the LRF methodology of Hassanzadeh and Kuang (2016) (see section 2d). Unlike a fluctuating EOF1, a steady EOF1 state enables the phase speed spectra to be easily calculated. The upper-level UV phase speed spectrum for the positive phase of EOF1 in the GCM is shown in Fig. 5a. UV increases at high phase speeds at and poleward of the mean jet (centered at ≈42°). There are two zones of weakening UV: one at low phase speeds centered on the mean jet and one in the subtropics at moderate phase speeds. This response looks similar to the response to decreases in zonal-mean component of friction in Lorenz (2015) except the second zone of decreasing UV is missing.

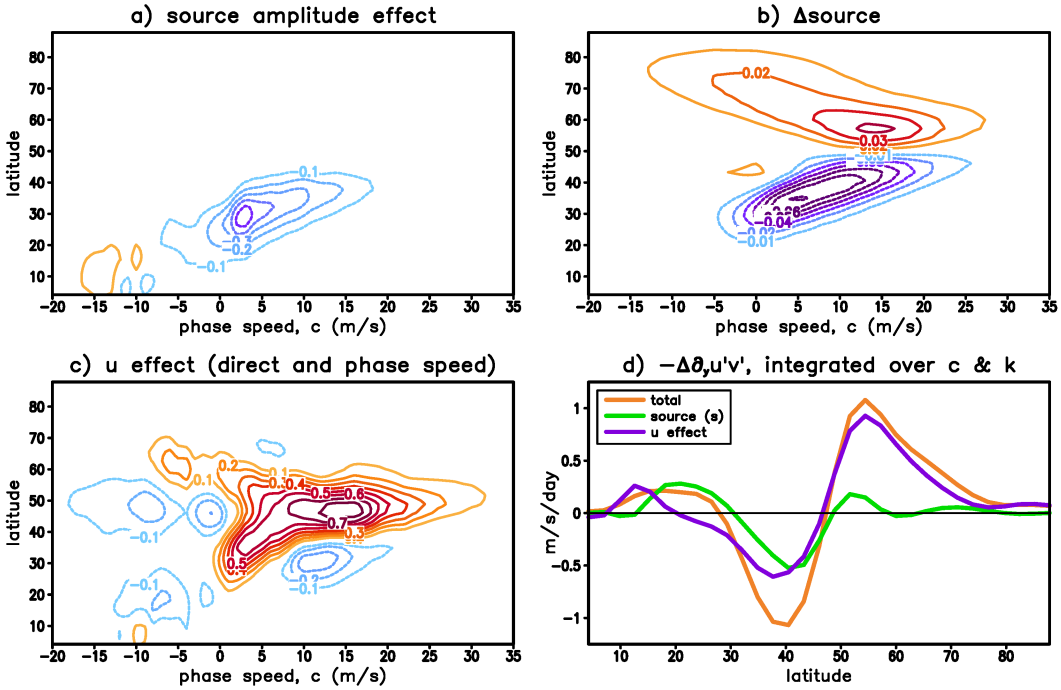

FIG. 6. Phase-speed/latitude spectrum of the response to EOF1 partitioned into various mechanisms using baroclinic RWC (see text). a) Change in $\overline{u'v'}$ due to baroclinic wave source amplitude. b) For reference, the change in baroclinic wave source. c) Change in $\overline{u'v'}$ due to background flow and baroclinic source phase speed effect. d) $\overline{u'v'}$ response in previous panels integrated over phase speed: total response from baroclinic RWC (orange), baroclinic source amplitude effect (green), background flow and baroclinic source phase speed effect (purple).

The barotropic RWC (Fig. 5b) gets the general pattern but (1) the increases in UV are too confined in latitude, (2) there are spurious decreases on the poleward side of the increases, (3) the decreases in the subtropics are too strong and (4) the decreases at low phase speed are too weak. The first two biases are also present in the response to decreases in zonal-mean component of friction seen in Lorenz (2015). The baroclinic RWC (Fig. 5c) improves the simulation of the regions of decreasing UV. In particular, the spurious region of decreasing UV on the poleward side is no longer present. The distribution of the UV increases are also more evenly distributed in latitude and phase speed, which is consistent with the GCM. Integrating the UV response over phase speed (Fig. 5d), we see that the baroclinic RWC improves the UV response at all latitudes. The main error in the baroclinic RWC is the overestimation of the response around 50°.

TABLE 1. Estimates of the relative contribution of barotropic and baroclinic mechanisms to the eddy feedback in the two-level GCM using different methods (see text).

Method	Experiment	Barotropic Feedback (%)	Baroclinic Feedback (%)
RWC	Baroclinic RWC	70.3	28.1
	Barotropic RWC	91.2	8.1
LRF EOF	Barotropic	62.8	37.2
	Baroclinic	71.5	28.5
	Strong Thermal Damping	77.0	23.0
	Upper Friction	87.1	12.9
	Force U only	70.1	29.9
IWP model		69.5	30.5

In Fig. 6, the baroclinic RWC is used to isolate mechanisms for the UV anomalies associated with EOF1 of U. The baroclinic feedback (Fig. 6a) explains a significant portion of the UV decreases in the subtropics at moderate phase speeds. The changes in baroclinic wave source that are driving the baroclinic feedback (Fig. 6b) are largest in the subtropics, which is consistent with Fig. 6a. However, there are also significant increases in source poleward of 50°. Apparently these sources are in regions of phase speed/latitude space that are unfavorable for Rossby wave propagation. The barotropic mechanism (background flow and phase speed changes, Fig. 6c) accounts for all of the increases in UV and a portion of the UV decreases. Fig. 6d shows the eddy momentum flux convergence, m, integrated over phase speed. We see that the m decreases seen equatorward of 45° are about equally split between barotropic and baroclinic mechanisms while the m increases further poleward are almost exclusively due to barotropic mechanisms. To get a single number quantifying the relative contribution of each mechanism on the total, we project each mechanism's m onto the total m and normalize by the total m amplitude ($= \sqrt{\sum_\phi m^2}$). According to this metric, the barotropic feedback contributes 70.3% to the eddy feedback and the baroclinic feedback contributes 28.1% (see first row of Table 1). We repeated this analysis but for barotropic RWC and the contribution of the barotropic feedback is even larger (Table 1). However, because the baroclinic RWC gives more realistic eddy fluxes, the baroclinic RWC results should be given more weight.

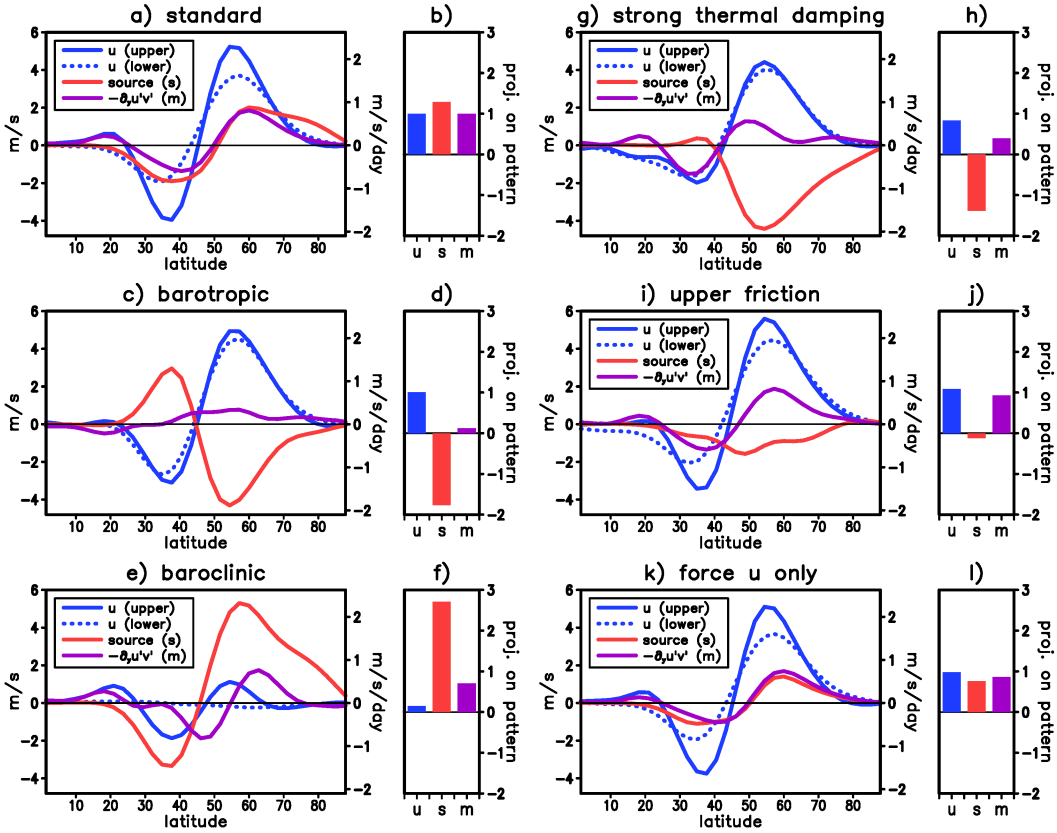

FIG. 7. Results of LRF experiments with prescribed EOF1 structures. a) Standard EOF1 U (blue, solid = upper-level, dashed = lower level). Response to standard EOF1 for baroclinic source (red) and eddy momentum flux convergence (purple). b) Blue: projection of U response onto the standard EOF1 (identically 1 in this case). Red: projection of baroclinic source response onto the standard eddy momentum flux convergence pattern. Purple: projection of eddy momentum flux convergence response onto the standard eddy momentum flux convergence pattern (identically 1 in this case). c) Same as (a) but for barotropic EOF1. d) Same as (b) but for barotropic EOF1 projected on standard patterns. e) and f) Same as (a) and (d) but for purely baroclinic EOF1. g) and h) Same as (a) and (d) but for strong thermal damping EOF1. i) and j) Same as (a) and (d) but for upper-level friction on U anomalies EOF1. k) and l) Same as (a) and (d) but for EOF1 forced via U instead of U and temperature.

c. Linear Response Function (LRF) Analysis

Hassanzadeh and Kuang (2016) developed a novel LRF methodology, which can be used to force arbitrary time- and zonal-mean U profiles in a GCM. For example, Hassanzadeh and Kuang (2019) used this methodology to impose a barotropic version of EOF1 in a GCM. In this paper, we implement the LRF methodology in the two level model and we manipulate the U EOF1 pattern in

order to understand the dynamics of the eddy feedback. First, we look at the LRF forced version of EOF1: positive phase minus negative phase (Fig. 7a). The upper and lower U (blue) are in phase but the lower U is smaller in amplitude, which implies the temperature gradient is colocated with the U anomalies. The eddy momentum flux convergence at the upper level (m, purple) is mostly in phase with U, consistent with the positive feedback, although m is displaced slightly poleward compared to U. The convergence of vertical EP flux in the upper level, or baroclinic wave source, (s, red) is aligned closely with m but of slightly larger amplitude. The fact that s is larger than m everywhere is consistent with the baroclinic feedback, although we will see that this is deceptive. For diagnosing the feedback mechanisms, we would like to quantify the U, m and s response with a single number. For U, we project the U (dependent on both latitude and pressure) on the standard EOF1 U and normalize by the standard U amplitude. By definition, this metric is one for the standard EOF (blue bar in Fig. 7b). Because s drives m according to the baroclinic feedback, s (dependent on latitude only) is projected on the standard EOF1 m and normalize by the standard m. Because s is projected on m, the amplitude is different than one (= 1.29) in the standard EOF case (red bar in Fig. 7b). Like U, which is projected on itself, m (dependent on latitude only) is projected on the standard m (purple bar in Fig. 7b).

For the next experiment, the target U anomaly is barotropic with U equal to the vertical average of the standard U. The resulting U is slightly off from completely barotropic (blue, Fig. 7c) but still very close. Consistent with Hassanzadeh and Kuang (2019), the m response disappears for barotropic EOF1 anomalies. Hassanzadeh and Kuang (2019) argued that this result is evidence that the baroclinic feedback is dominant. However, barotropic U do not give zero s, instead s is strongly opposite U. Currently, we do not understand *why* s is opposite U. But if one accepts that *both* baroclinic and barotropic feedbacks are operating, then this experiment provides a way to disentangle their relative contributions. For example, the s projection decreases from 1.29 in Fig. 7b to -1.77 in Fig. 7d. Meanwhile, m decreases from 1.0 to 0.12 and U hardly changes. Therefore, a 3.06 decrease in s leads to a 0.88 decrease in m, which implies that 1 unit of s leads to 0.29 ($= 0.88/3.06$) units of m. Because the s amplitude in the standard EOF case is 1.29, the fraction of m due to the baroclinic feedback in the standard case is 0.37 ($= 1.29 \cdot 0.29$). More precisely, with two simulations, we can find the two coefficients that determine the strength of the barotropic and

baroclinic feedback:

$$
\begin{aligned}
a_U U_0 + a_s s_0 &= m_0, \\
a_U U_1 + a_s s_1 &= m_1,
\end{aligned}
\tag{9}
$$

where U, s and m are the projection values in the bar chart in Fig. 7, the subscripts 0 and 1 refer to the standard EOF1 and alternate EOF1, respectively, and a_U and a_s are constant coefficients that are calculated from the simple system of two equations. After the coefficients are determined, $a_U U_0/m_0$ and $a_s s_0/m_0$ are the relative contribution of the barotropic and baroclinic feedback for the standard EOF1. Similar expressions for the non-standard EOF structure involve subscripts of 1 instead of 0. The estimates for the standard EOF, which are given in Table 1, agree with the simple estimate in this particular case. Although Hassanzadeh and Kuang (2019) did not analyze the EP flux budget for their barotropic EOF, their heat fluxes were the opposite sign as U, which is consistent with our results.

To test the robustness of the above estimate, we can also impose alternative U anomalies. In Fig. 7ef, the baroclinic portion of EOF1 is imposed: the lower level U is zero and the upper level U is the difference between upper and lower in the standard case. In this baroclinic EOF case, the barotropic feedback is small (blue bar in Fig. 7f), so almost all the feedback is baroclinic. Apparently, for an almost pure baroclinic feedback, s leads to m anomalies that are about four times smaller (compare red and purple bars in Fig. 7d). The estimates for the barotropic and baroclinic feedback for the standard EOF from this case are 71.5% and 28.5%, respectively, which is comparable to the previous case.

An alternate way to generate a barotropic EOF is to strongly damp zonal-mean temperature anomalies back to climatology. In contrast to imposing a barotropic version of EOF1, the eddies have some control over the shape of the EOF. This experiment is inspired by similar experiments in Zurita-Gotor et al. (2014). For this experiment, the damping time scale is set to 0.1 days^{-1} and the leading EOF1 structure was calculated from model simulation. Next, we imposed this alternate EOF structure in the control simulation using the LRF methodology (Fig. 7g). The U response is nearly barotropic like the previous barotropic experiment (Fig. 7c), however, the equatorward center of action extends to lower latitudes in the strong thermal damping experiment. The U, s and m projections in the new experiment are similar to the barotropic experiment and the estimates

for the barotropic and baroclinic feedback for the standard EOF from this case are 77.0% and 23.0%, respectively (Table 1). Like the barotropic case, we believe the correct interpretation of the strong thermal damping case depends critically on recognizing the large negative baroclinic source feedback that accompanies barotropic U anomalies. Assuming the s response is negligible when U is barotropic leads to the erroneous conclusion that the barotropic feedback is negligible (Zurita-Gotor et al. (2014); Hassanzadeh and Kuang (2019)).

In Fig. 7i, we repeat the upper friction experiment from Section 3a in the two level model, and then prescribe EOF1 from this simulation using the LRF methodology. As in the multi-level model, the s response in this simulation is small and therefore the estimate of baroclinic feedback is especially small for this case (12.9%). We do not understand why the estimate is lower in this case but it is interesting that the two cases where the EOF structure is determined by the GCM (Upper Friction and Strong Thermal Damping) have smaller baroclinic feedback estimates than the two cases where the EOF structure is simply prescribed (Barotropic and Baroclinic).

Most of the above experiments involve dramatic changes in EOF1 structure. In the final experiment, we try a subtle change in EOF structure. Hassanzadeh and Kuang (2016) develop the LRF matrix by forcing both U and temperature. All the experiments above follow this convention. In the last experiment, we develop a separate LRF matrix that involves the forcing of U only. The forced EOF1 structure in the experiment (Fig. 7k) is very close to the standard experiment that forces both U and temperature (Fig. 7a). The differences between the experiments is more evident in the relative amplitude of s and m. Using these two experiments to constrain the two feedback mechanisms (9) gives 70.1% and 29.9% for the barotropic and baroclinic feedbacks, respectively. This estimate is the median of all LRF experiments and is very close to the Baroclinic RWC estimate. It is possible that the estimates from this experiment should carry more weight because the EOF structure is not much different than the control, however this is just conjecture.

d. Inferred Wave Propagation (IWP) Model Based on Mean State

For our final estimates, we use the meridional wave propagation implied by the time-mean EP flux budget to create a model that predicts the meridional wave activity flux (= negative eddy momentum flux) from the wave source. We call this simple model the Inferred Wave Propagation (IWP) model. When applied to a perturbed simulation (i.e. provide perturbed wave source as

input), the model gives the meridional wave activity flux changes due to source changes in the absence of any changes in the mean wave propagation characteristics. In other words, the model gives an estimate the baroclinic feedback. Explicitly, the poleward and equatorward wave activity flux in the IWP model are given by:

$$\begin{aligned} \frac{dF_n}{d\phi} &= w_n s - DF_n, \\ \frac{dF_s}{d\phi} &= w_s s + DF_s, \end{aligned} \tag{10}$$

where F is the meridional wave activity flux, the subscripts n and s refer to northward and southward propagating waves, s is the baroclinic wave activity source, D is the dissipation, w_n and w_s are the fraction of wave activity propagating northward or southward, respectively, from the wave source ($w_n + w_s = 1$) and ϕ is the latitude in radians. The northward equation is integrated from south to north starting with $F_n = 0$ at the south-pole, and describes the accumulation of wave activity flux $w_n s$ and linear dissipation of wave activity. The southward equation is integrated from north to south starting with $F_s = 0$. Due to the direction of integration in (10), $F_n >= 0 >= F_s$. The boundary conditions implicitly assume no wave reflection at the poles. The total wave activity flux, F, is

$$F = F_s + F_n. \tag{11}$$

Equation (10) is first applied to the mean state in an inverse sense: D and w are calculated from the time-mean s and F. Both D and w are assumed to vary with latitude. Because there are two knowns (F and s) but three unknowns (F_n, F_s and either w_s or w_n), the problem is under-constrained. Here we consider the simplest case, which has wave-propagation-direction weights given by the sign of F:

$$\begin{aligned} &w_n = 1,\ w_s = 0,\ \text{if}\, F > 0 \\ &w_n = 0,\ w_s = 1,\ \text{otherwise.} \end{aligned} \tag{12}$$

Consider a latitude zone where $F > 0$ and $w_n = 1$. If we use the approximation $F_n = F$ within this zone, then the F_n equation becomes:

$$\frac{dF}{d\phi} = s - DF. \tag{13}$$

The divergence of the EP flux, E, is by definition $-s+\frac{dF}{d\phi}$. Therefore a simple manipulation of (13) implies that the dissipation is given by: $D=-\frac{E}{F}$. Note that D is positive because E is negative. When $F<0$ and $w_s=1$, the same manipulation of terms gives $D=+\frac{E}{F}$. Generalizing for both cases,

$$D=\left|\frac{E}{F}\right|. \tag{14}$$

Once D, w_n and w_s are known, the wave activity flux (and eddy momentum flux) can be found by integrating each component of (10) over latitude starting at the appropriate pole. At some latitudes, D can be quite large, which causes standard explicit schemes to be unstable when integrating on the GCM latitude grid. Since (10) is linear, the easiest approach is to use absolutely stable implicit schemes. Here we use the 4th order Backward Differentiation Formula (BDF) (Curtiss and Hirschfelder (1952); Süli and Mayers (2003)). For initializing, we incrementally use with the 1st, 2nd and 3rd order BDF schemes.

To apply (10) to the spherical geometry, F, s and E should be scaled by $\cos\phi$ as follows:

$$\begin{aligned} F &= -\overline{u'v'}\cos^2\phi \\ s &= -\cos\phi\int_{500\text{mb}}^{100\text{mb}}\frac{\partial F_p}{\partial p}\partial p \\ E &= \cos\phi\nabla\cdot\mathbf{F}, \end{aligned} \tag{15}$$

where $\mathbf{F}$ is the two dimensional, spherical-geometry EP flux vector given in Edmon et al. (1980), F_p is the vertical EP flux vector in Edmon et al. (1980) and the integral in the s equation is over the upper troposphere.

The IWP model is first applied to the upper level of the two-level primitive equation model. Because the model is trained on the mean state, the time-mean EP flux divergence is very well captured (Fig. 8a, compare red line to dashed purple line). This result simply validates the assumption that $F=F_n$ when $w_n=1$ and $F=F_s$ when $w_s=1$, which is used in the derivation of D in (14). The reciprocal of D (Fig. 8b) maximizes in the subtropics around 30°, which is the same latitude that the wave source, s, and the negative EP flux divergence, $-E$, cross in Fig. 8a. This is consistent with the fact that a strong wave activity flux is required here to transport from latitudes where $s>-E$ to latitudes where $s<-E$. In the polar regions, on the other hand, the dissipation

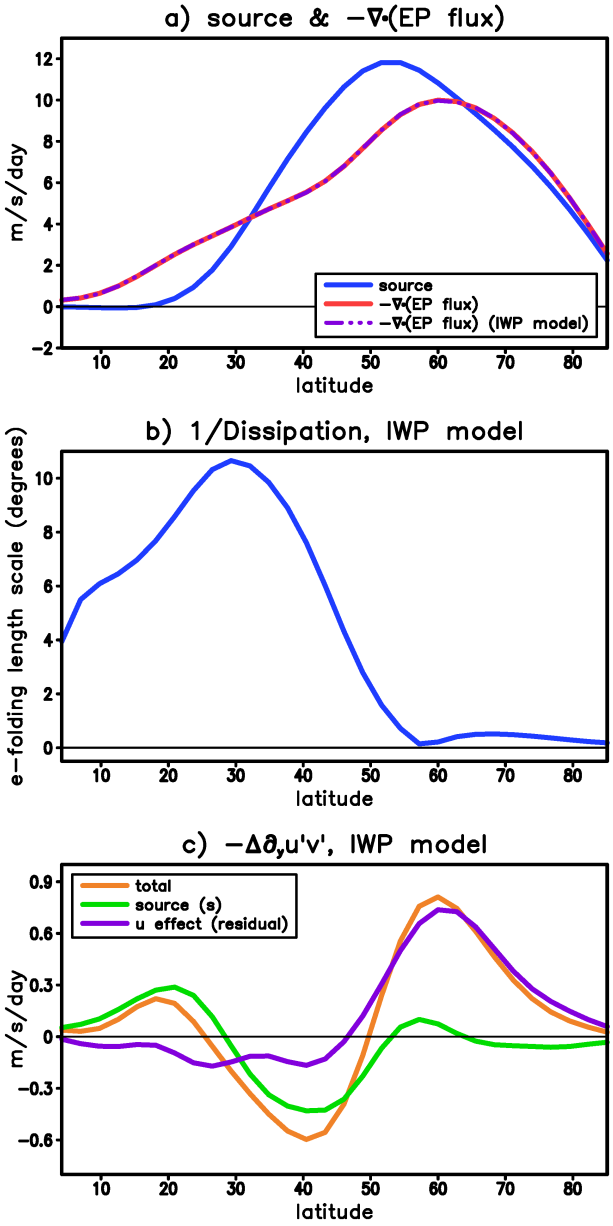

FIG. 8. a) Time-mean baroclinic source (blue) and EP flux convergence (red) at upper-level in two-level model. EP flux convergence from IWP model (purple dashed). b) Reciprocal of dissipation in IWP model (degrees). c) Mechanisms of eddy momentum flux response to EOF1 from IWP model: total (orange), baroclinic source (green), effect of background flow (purple).

e-folding scale $(1/D)$ is very small, which is consistent with the fact that little wave propagation is needed because $s \approx -E$

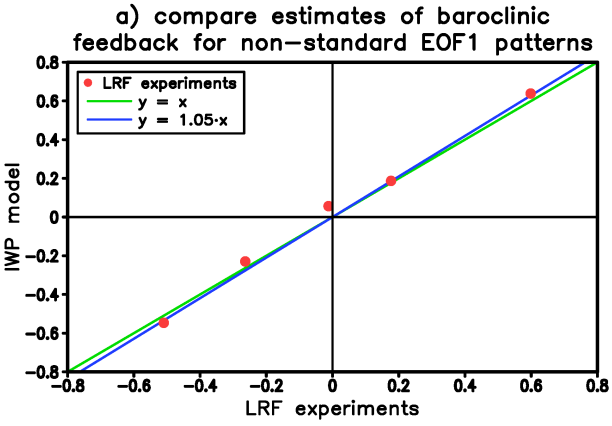

FIG. 9. Scatter plot of magnitude of the baroclinic feedback associated with the non-standard EOF structures in Fig. 7: x-axis: estimate from the LRF experiments using (9), y-axis: estimate from the IWP model. The line $y = x$ is green, the least squares line, $y = 1.05 \cdot x$ is blue

Fig. 8c, we show the change in eddy momentum flux convergence, m, in response to the EOF1 from the LRF simulations (orange line). To estimate the baroclinic feedback, the change in s is input to the IWP model to estimate m in the absense of changes in wave propagation (i.e. D, w_n and w_s are unchanged). This m from the baroclinic feedback (green) is remarkably similar to the baroclinic feedback estimated from RWC (Fig. 6d). In both cases, the baroclinic feedback is very weak for the 60° center of action, maximizes at the 40° center of action and dominates the changes deep in the subtropics. The barotropic feedback (purple line), which is simply the residual in this case, dominates the m response for the 60° center of action. The relative contribution of the barotropic and baroclinic mechanisms to the total m response is 69.5% and 30.5%, respectively (Table 1). This agrees with our previous estimates.

The IWP model can also be compared with the LRF results for the baroclinic feedback for the non-standard EOF1 structures. For example, the baroclinic feedback for the strong thermal damping EOF1 is large and negative (Fig. 7gh). In Fig. 9, we show the amplitude of the baroclinic feedback for the five non-standard EOF structures in Fig. 7 for the IWP model versus the LRF estimates (see section 3c). The close correspondence suggests that the IWP model is in fact able to separate the contributions of the baroclinic and barotropic feedback.

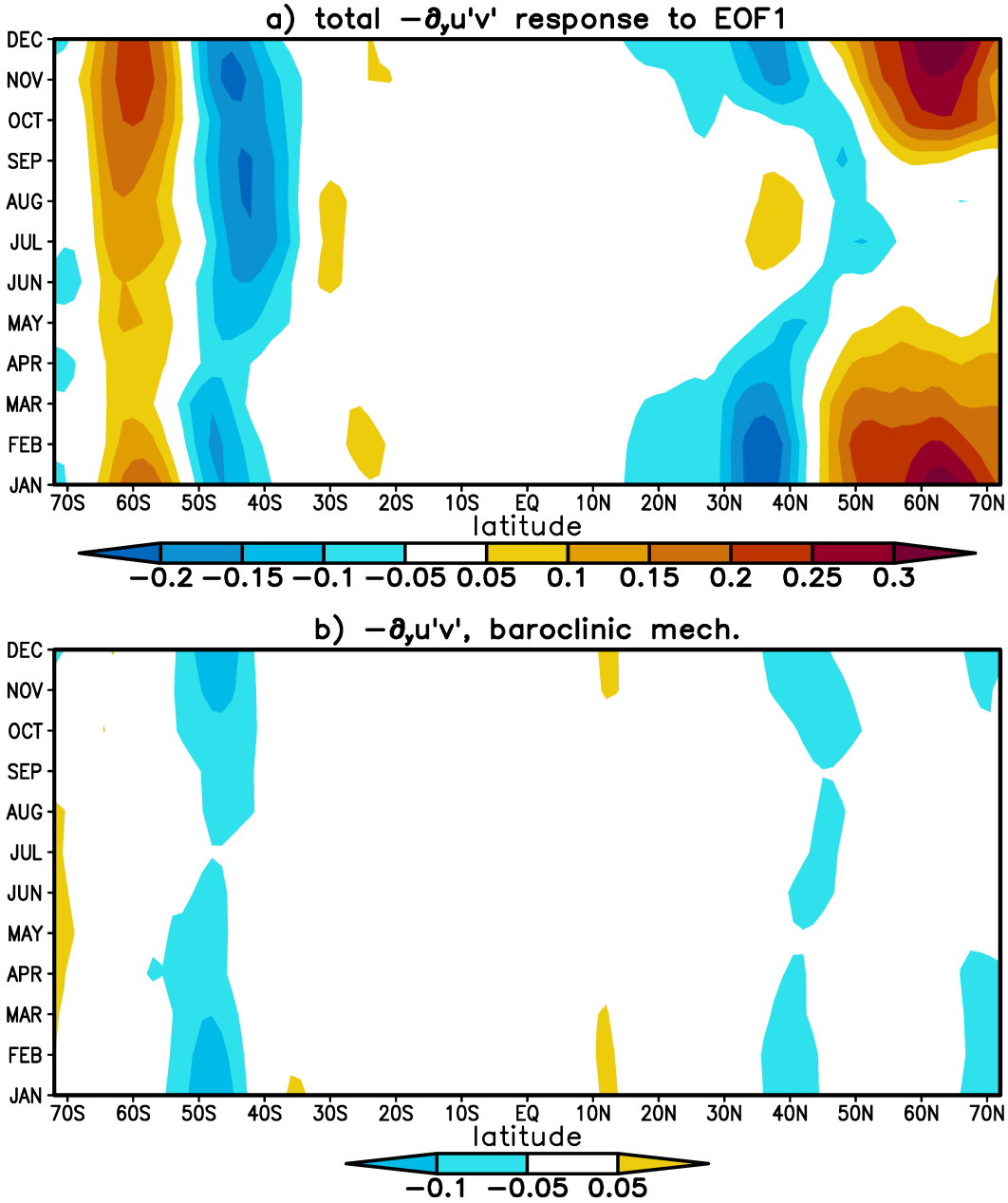

FIG. 10. a) Response of eddy momentum flux convergence to EOF1 estimated from lagged regressions (m/s/day). x-axis is latitude and y-axis is month. The panel shows results of two separate lagged regressions-one for the Southern Hemisphere and one for the Northern Hemisphere-that are joined together at the equator. b) Same as (a) but for the portion of the eddy momentum flux convergence from the baroclinic feedback (estimated from the IWP model).

4. Baroclinic Feedback Estimates from Observations

In this section, we apply the IWP model to ERA-interim reanalysis (Dee et al. (2011)) data to estimate observed feedbacks. The mean EP fluxes from the IWP model deviate from observed EP fluxes in the deep tropics due to tropical Rossby wave sources. In the extratropics, however, the model captures the observed EP fluxes.[1] To estimate the eddy fluxes "caused" (however, see Byrne et al. (2016)) by U anomalies, we first use time-lagged regression analysis between the leading U PC and eddy wave activity fluxes (Lorenz and Hartmann (2001)). Next, the eddy fluxes are averaged from lags 10-15 days after the peak in the U PC. Following the RWC methodology, the EP flux budget is then averaged from 500-100mb.

In Fig. 10a, we show the eddy momentum flux convergence, m, response to EOF1 as a function of time of year (5-month "seasons" centered on each month). Two analyses, one for each hemisphere, are joined at the equator and superimposed on the same plot. The response is largest in Northern Hemisphere winter, consistent with Lorenz and Hartmann (2003), and smallest in Northern Hemisphere winter. In the Southern Hemisphere, the m response is largest in November and December, which is consistent with the seasonal cycle of annular mode persistence found in Baldwin et al. (2003).

The baroclinic feedback estimated from the IWP model is shown in Fig. 10b. Like the two-level GCM (Fig. 8c), the baroclinic feedback accounts for a portion of the equatorward center of action of m and practically none of the poleward center of action. In the Northern Hemisphere, the latitude of the baroclinic feedback varies much less through the seasonal cycle than the total m.

Projecting the baroclinic feedback on the total m and normalizing by the m amplitude gives its relative contribution to the total (Fig. 11). The barotropic feedback is defined as the residual. During some time periods, for example Northern Hemisphere winter, the contribution of the baroclinic feedback is zero. Evidently, the baroclinic feedback structure is orthogonal to the total m. The baroclinic feedback makes the largest contribution to m in March in the Southern Hemisphere and June in the Northern hemisphere where it approaches or exceeds 40% of the total. In most cases, however, the baroclinic feedback in observations is weaker than that in the two-level primitive equation model.

[1]We have also attempted to apply RWC to observations, however, the simulation of the climatological UV is poor, perhaps due to zonal asymmetries in the U climatology.

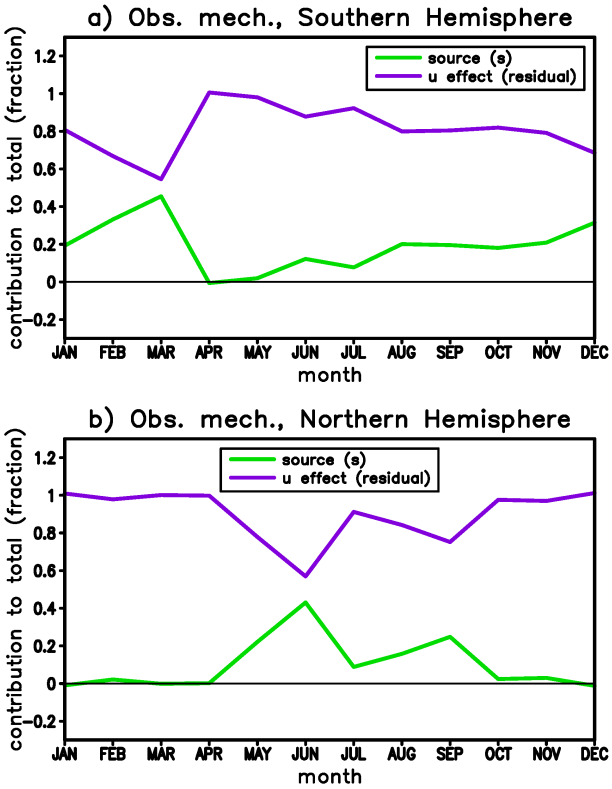

FIG. 11. Contribution of the baroclinic feedback (green) to the total eddy momentum flux convergence response as a function of month. The residual is the estimate for the background flow effect (purple).

5. Conclusions

In this paper, we provide multiple independent estimates of the relative role of baroclinic and barotropic feedbacks on the eddy response to EOF1 U anomalies. All estimates suggest that barotropic feedbacks are the most important (70% of total) but that baroclinic feedbacks are non-negligible (30%). The estimates come from the following methods/experiments: Rossby Wave Chromatography (RWC, Lorenz (2015) and Section 2c), Linear Response Function (LRF) methodology (Hassanzadeh and Kuang (2016)) and a new method we call Inferred Wave Propagation (IWP) modelling.

Using linearized dynamics, RWC calculates the eddy momentum flux from 1) the baroclinic wave activity source and 2) the meridional wave propagation characteristics of the background flow. By changing these two components separately, the relative roles of the baroclinic (component 1) and barotropic (component 2) feedback on U anomalies. The structure of the momentum flux response to each component can also be calculated.

Hassanzadeh and Kuang (2019) argued for baroclinic mechanisms by manipulating the vertical structure of EOF1 using the LRF method. They found that a barotropic version of EOF1 did not have a positive eddy feedback, presumably because the baroclinic source does not change for barotropic anomalies. We repeat these experiments, however, and find that the baroclinic source response is not zero but strongly negative, damping the eddy feedback. Here, we start with the hypothesis that both the baroclinic and barotropic mechanisms are relavant. With two mechanisms and two cases (standard EOF and barotropic EOF) one can linearly disentangle the role of each mechanism. Consistent with RWC, we find that the barotropic mechanism is more important. We repeat this analysis for multiple alternative changes in the EOF1 vertical structure: for example, baroclinic U anomalies with zero surface U. All five perturbed EOF1 structures imply similar magnitudes of the barotropic and baroclinic feedback.

The LRF experiments also show that barotropic U anomalies lead to opposing, negative baroclinic source anomalies. Currently, we do not understand the dynamical reasons for such a response although one potential explanation involves Gliatto and Held (2020), who find that Rossby waves propagating meridionally in a vertically and meridionally varying background flow can give rise to non-zero vertical EP flux.

The final estimate of the relative role of baroclinic and barotropic feedbacks uses the upper-level wave activity budget of the mean state to quantitatively infer the degree of meridional wave propagation in the mean state. This method, which we call Inferred Wave Propagation (IWP) modelling, predicts the momentum flux given the baroclinic wave activity source assuming wave propagation characteristics are the same as the mean state. When applied to an anomalous wave source, IWP gives the momentum flux from the baroclinic feedback. IWP more precisely quantifies the ideas on "radiation efficiency" introduced by Robert et al. (2017). The baroclinic/barotropic feedback estimates from IWP agree with RWC and LRF and moreover, the spatial structures of the two feedback components are very similar to the baroclinic RWC method. IWP can also be applied to feedbacks diagnosed from observations using lagged regression analysis (e.g. Lorenz and Hartmann (2001)). Most times of year, the barotropic feedback is greater than or equal to the value from the simple GCM (70%) and in some cases the barotropic feedback accounts for almost all of the feedback. During March in the Southern Hemisphere and June in the Northern Hemisphere the barotropic feedback decreases to 60% of the total feedback.

All diagnostics and experiments in this paper suggest that barotropic feedbacks are dominant over baroclinic feedbacks. This is consistent with the success of barotropic models in capturing the structures and feedbacks associated with EOF1 U anomalies (Jin et al. (2006a); Jin et al. (2006b); Barnes and Hartmann (2011); Lorenz (2014b)).

Acknowledgments. This research was supported by NSF grant AGS-1557353.

Data availability statement. Please contact the author for the model code and data.

APPENDIX

Baroclinic RWC Methodology

In this appendix, we describe the methodology for prescribing the vertical EP flux in the linearized two-level QG model. First, the solutions to (5) are separated into individual zonal wavenumbers and phase speeds. Next, each of the resulting linear systems are written in matrix form:

$$\boldsymbol{\psi} = \mathbf{AF}, \tag{A1}$$

where $\mathbf{A}$ is a matrix, $\boldsymbol{\psi}$ and $\mathbf{F}$ are vectors of complex streamfunction and forcing amplitudes and we use simple second order differencing for the numerics. The vectors $\boldsymbol{\psi}$ and $\mathbf{F}$ are written with all latitudes of level 1 first and then all latitudes of level 2. As discussed in Lorenz (2015), using a vector to represent F assumes that F is coherent at all latitudes (i.e., at all times, F at high latitudes always has the same phase and amplitude relationship to F at low latitudes). Therefore, we follow Lorenz (2015) by adding a second dimension to F, which represents independent "realizations" of F. The individual realizations are analogous to the partitioning of data into separate chunks when doing cross-spectral analysis. The same realization argument holds for the streamfunction, $\boldsymbol{\psi}$. Therefore, (A1) becomes a matrix equation:

$$\mathbf{\Psi} = \mathbf{AF}. \tag{A2}$$

Multiplying each side of (A2) by the complex conjugate transpose of itself:

$$\mathbf{\Psi} = \mathbf{ACA}^H, \tag{A3}$$

where $\mathbf{\Psi}$ and $\mathbf{C}$ are the ψ and F covariance matrices, respectively, and $\mathbf{A}^H$ denotes the conjugate transpose. We assume that $\mathbf{C}$ is real, which means that, on average, there is no net horizontal- or vertical-phase tilt to the nonlinear forcing. Let n be the number of latitudes, then the covariance between the upper- and lower-level streamfunction is $\Psi_{j,n+j}$, where j is an index over latitude. The

vertical EP flux is proportional to the imaginary part of $\Psi_{j,n+j}$:

$$\text{source} = \frac{mk_D^2}{2a}\Im(\Psi_{j,n+j}), \tag{A4}$$

where m is the integer zonal wavenumber and the factor of two comes from the fact that the zonal mean of two waves with complex amplitudes x and y is $xy^*/2$. Equation (A4) gives n constraints on the solution to (A3). In addition, we constrain the ratio of upper- to lower-level streamfunction amplitude:

$$\sum_j \left(p_{\text{upper}}\cos\phi\right)^2 = A\sum_j \left(p_{\text{lower}}\cos\phi\right)^2, \tag{A5}$$

where A is the ratio from the GCM and j is an index over latitude. There are a total of $n+1$ contraints. The problem is to find either $\mathbf{\Psi}$ or $\mathbf{C}$ from these contraints. Because of the assumption that $\mathbf{C}$ is real, it is most straightforward to treat $\mathbf{C}$ as the unknown variable subject to the $n+1$ contraints. Because $\mathbf{C}$ is symmetric, there are $\frac{n(n+1)}{2}$ unknowns. To close the problem, we choose the smallest forcing (= sum of diagonal elements of $\mathbf{C}$) that satisfies the constraints and is a valid covariance matrix (i.e. $\mathbf{C}$ is positive semidefinite). This is a "semidefinite programming" problem (Helmberg et al. (1996); Vandenberghe and Boyd (1996)) and multiple high quality libraries exist in multiple popular programming languages. Once the solution ($\mathbf{C}$) is found, the streamfunction is given by the diagonal elements of $\mathbf{\Psi}$ in (A3). To find the eddy momentum flux, one first creates a matrix that converts streamfunction to v and a matrix that converts streamfunction to u. One then right multiplies $\mathbf{\Psi}$ by the v matrix and left multiplies $\mathbf{\Psi}$ by the complex conjugate transpose of the u matrix. The eddy momentum flux is the $\frac{1}{2}$ times the real part of the diagonal elements of the resulting matrix.

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